



Multimodal Machine Learning

Lecture 6.1: Multimodal Transformers (Part 2)

Dam Quang Tuan

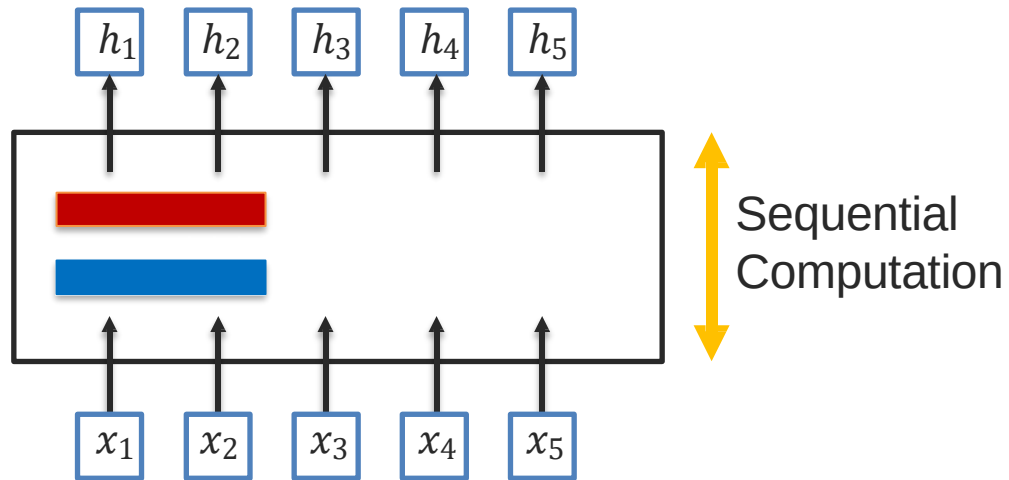
Objectives of today's class

- Visual transformers
 - Vision transformer (ViT)
 - Masked Auto-Encoder (MAE)
- Visual-language transformers:
 - ViLT = ViT+BERT
 - Vision-Language Caption (MAE+BERT)
- Video transformers

Vision Transformers

Recap: CNNs vs Transformers

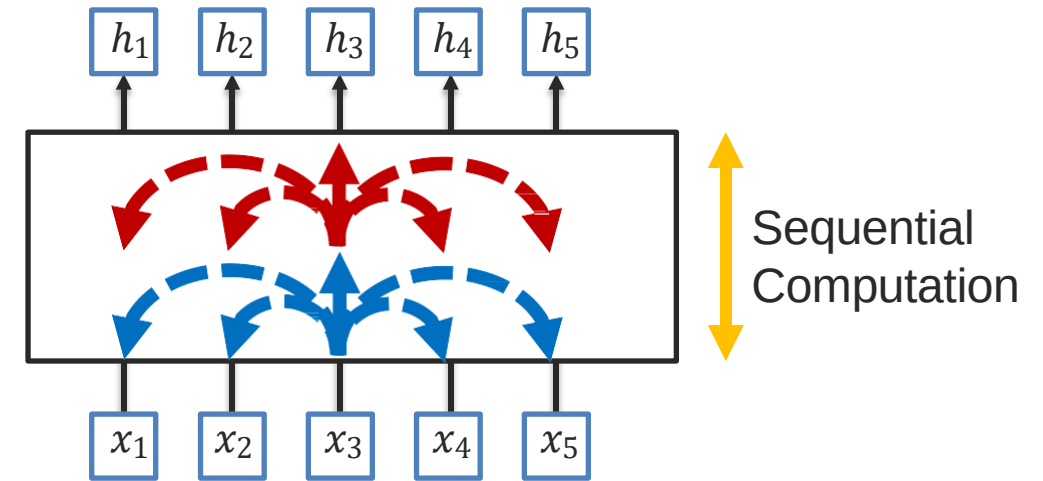
Convolutions



Can be parallelized!

But modeling long-range dependencies requires many layers.
And convolutional kernels are static.

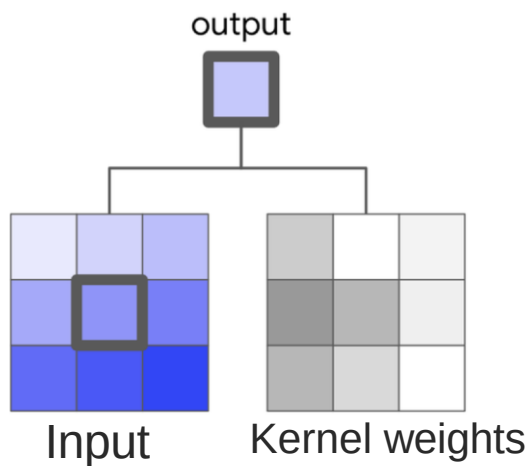
Self-attention



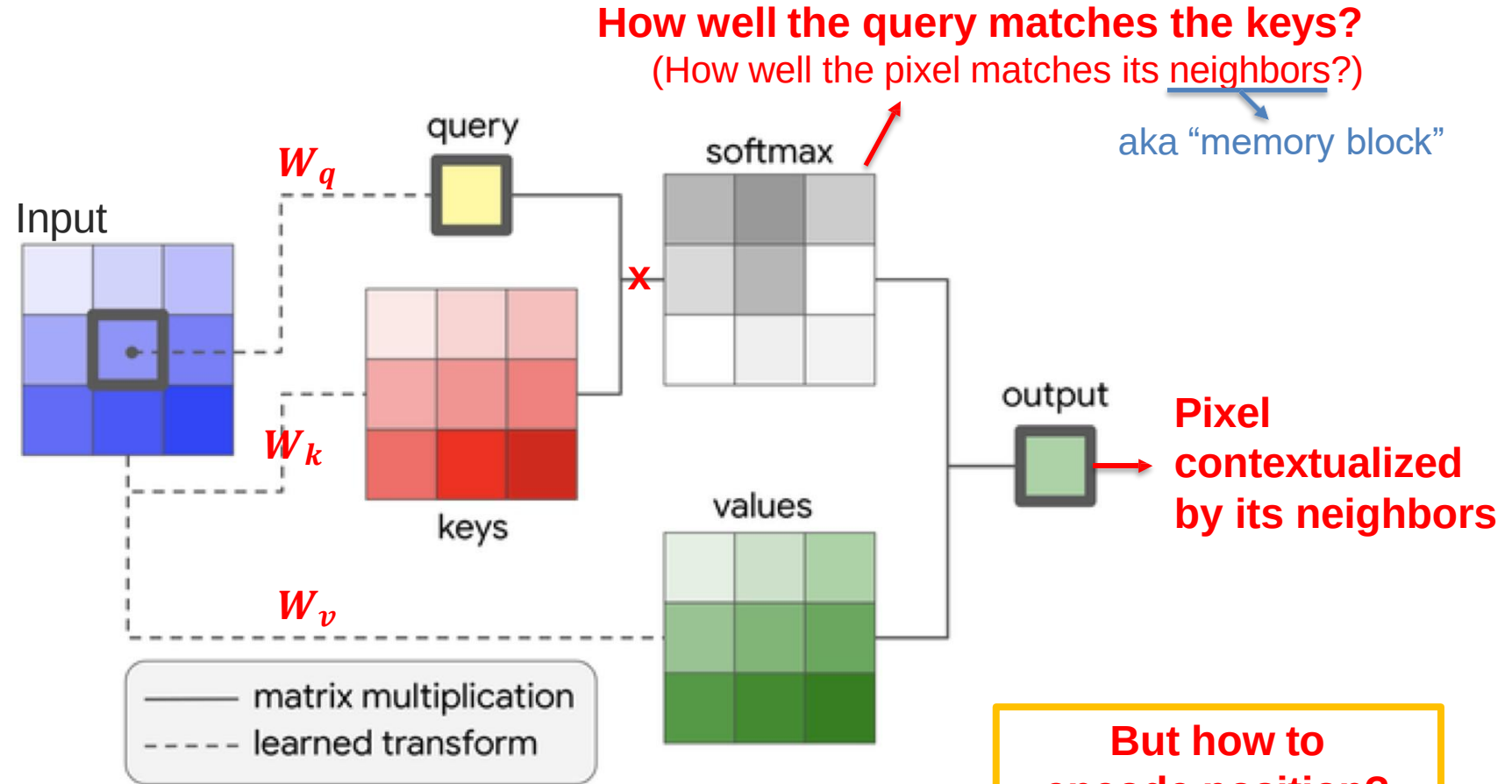
Can be parallelized!

Long-range dependencies
Dynamic attention weights
No inductive bias toward locality

Replacing a CNN w/ Self-Attention



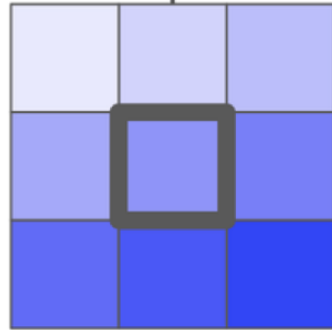
Convolution



Self-attention

Replacing a CNN w/ Self-Attention

Image patch



2D relative
position
embedding

-1, -1	-1, 0	-1, 1	-1, 2
0, -1	0, 0	0, 1	0, 2
1, -1	1, 0	1, 1	1, 2
2, -1	2, 0	2, 1	2, 2

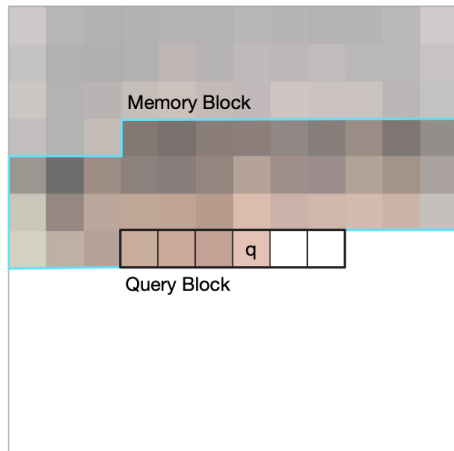
Position embedding is added to the key:

$$y_{ij} = \sum_{a,b \in \mathcal{N}_k(i,j)} \text{softmax}_{ab} \left(q_{ij}^\top k_{ab} + q_{ij}^\top r_{a-i,b-j} \right) v_{ab}$$

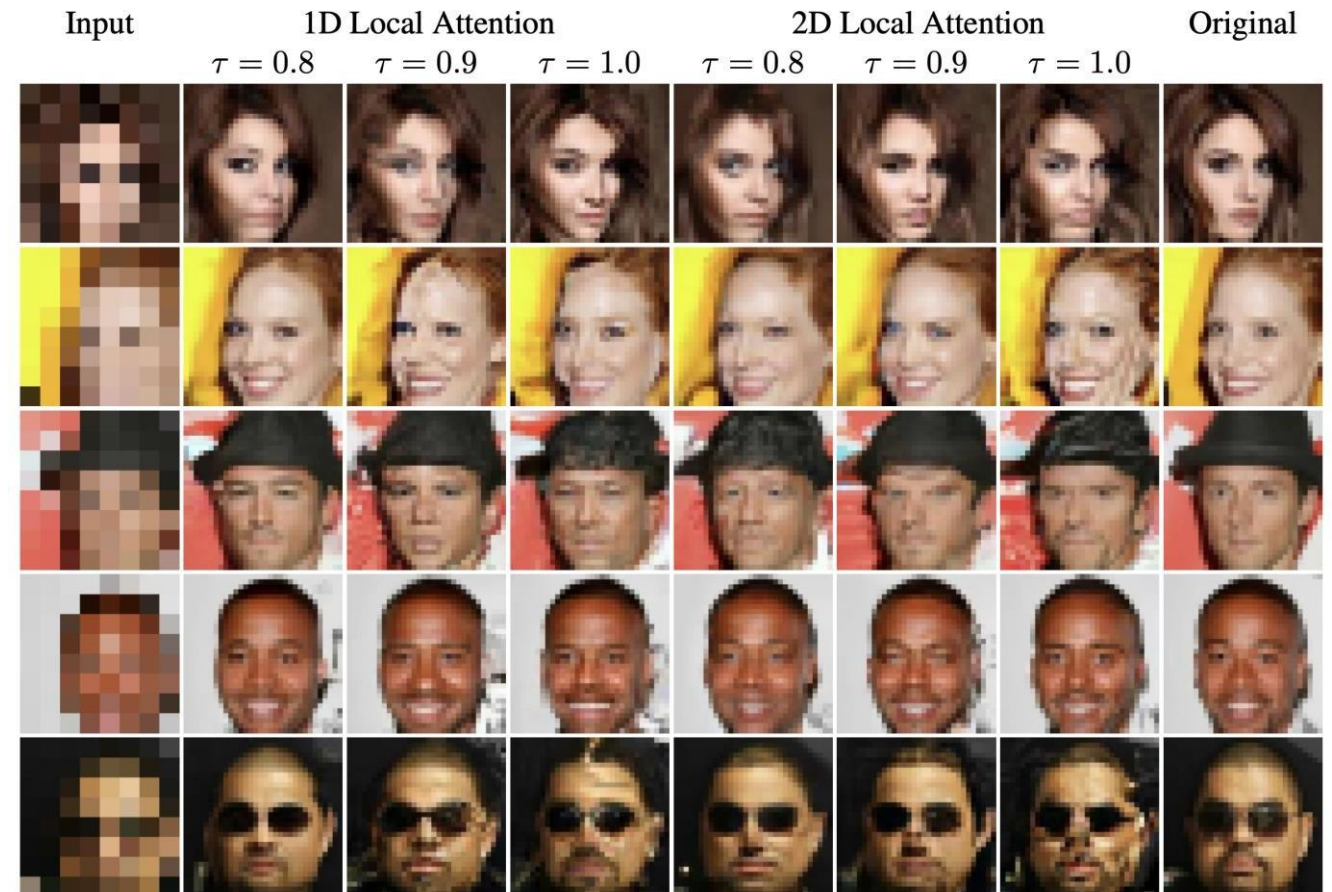
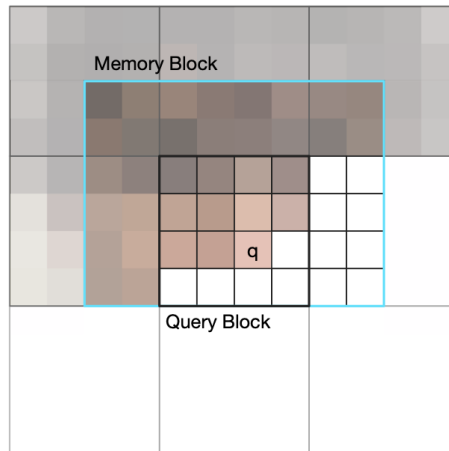
Pixel-Based Image Generation via Transformers

Produce 32x32 images, one channel of each pixel at a time. $3 \times 32 \times 32 = 3072$ positions

Local 1D Attention



Local 2D Attention

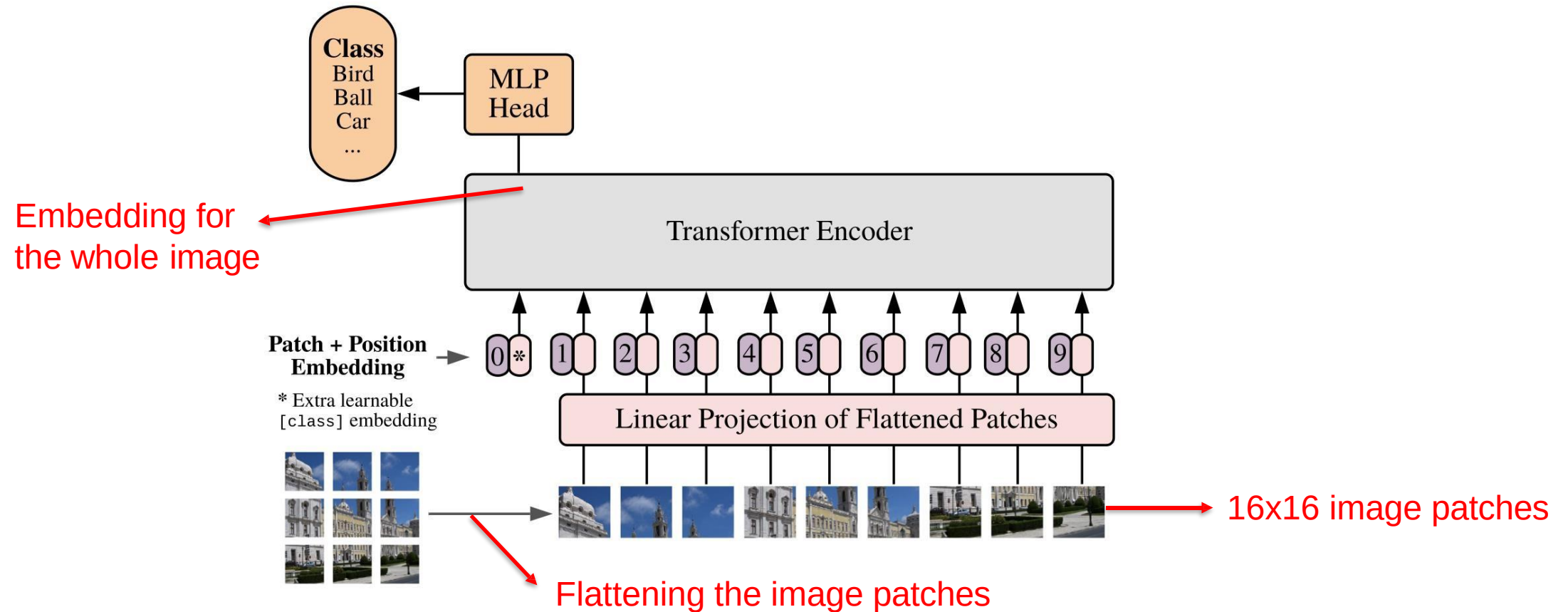


Vision Transformer (ViT)



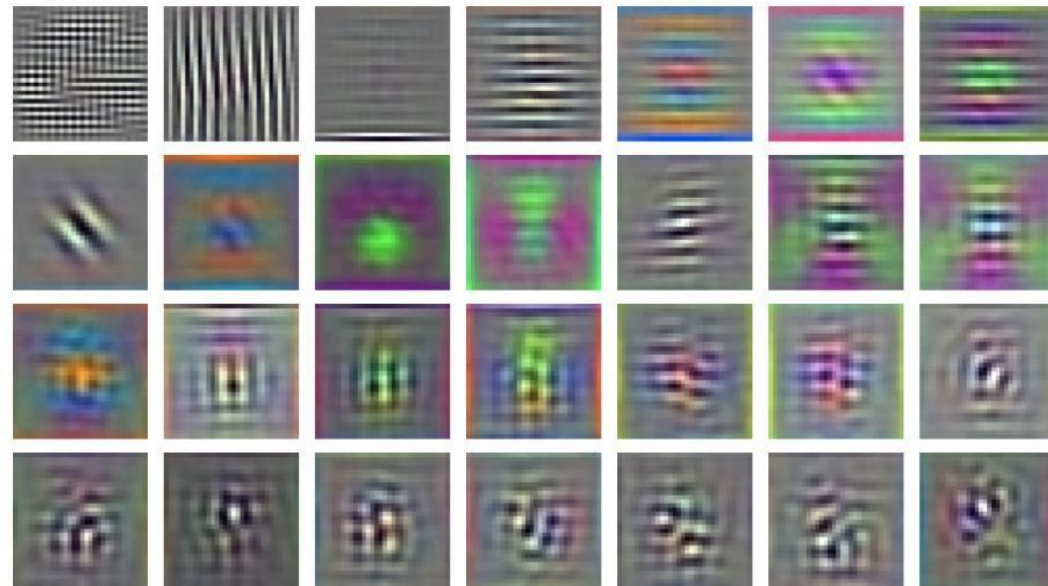
Dosovitskiy, Alexey, et al. "An image is worth 16x16 words: Transformers for image recognition at scale." *arXiv* (2020).

Vision Transformer (ViT)

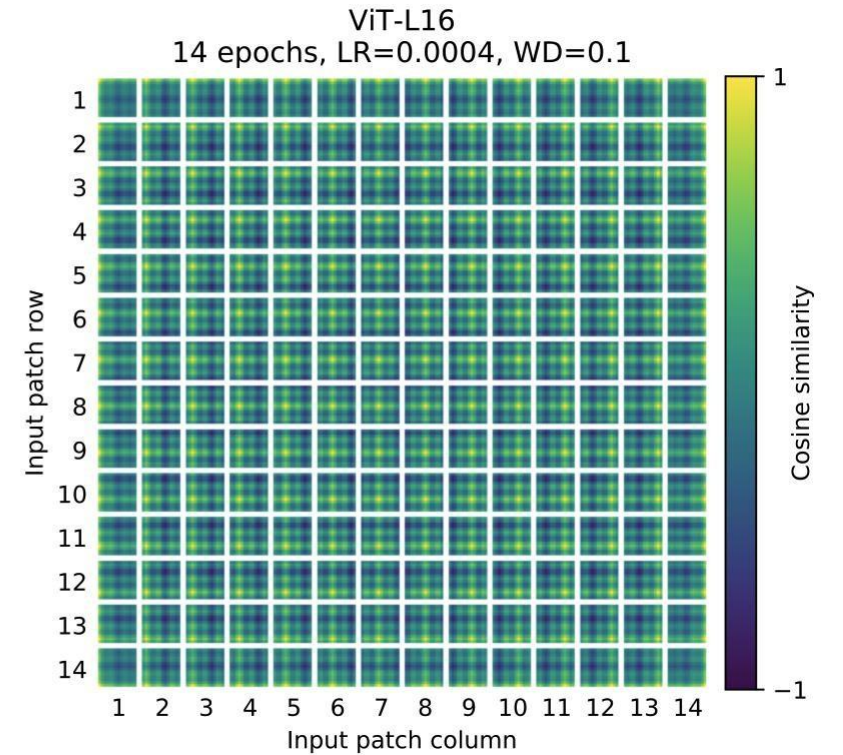
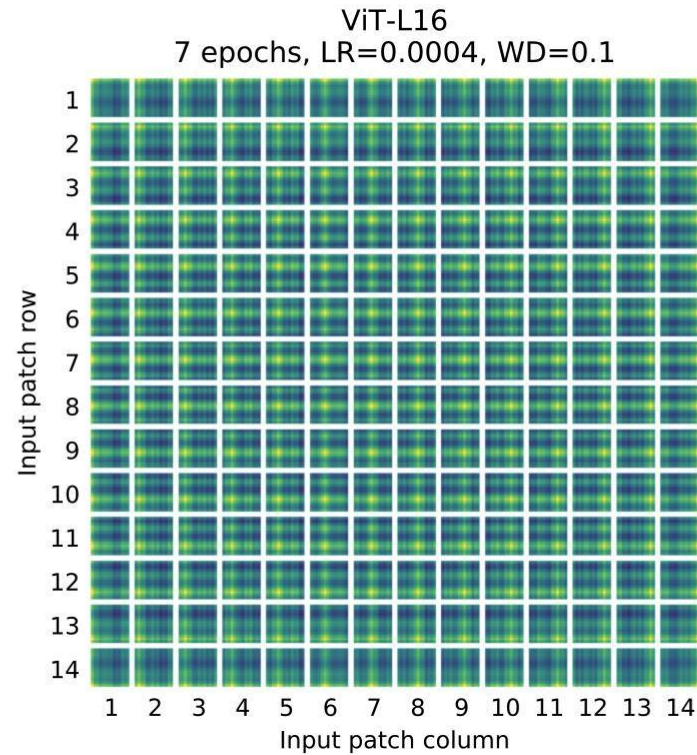
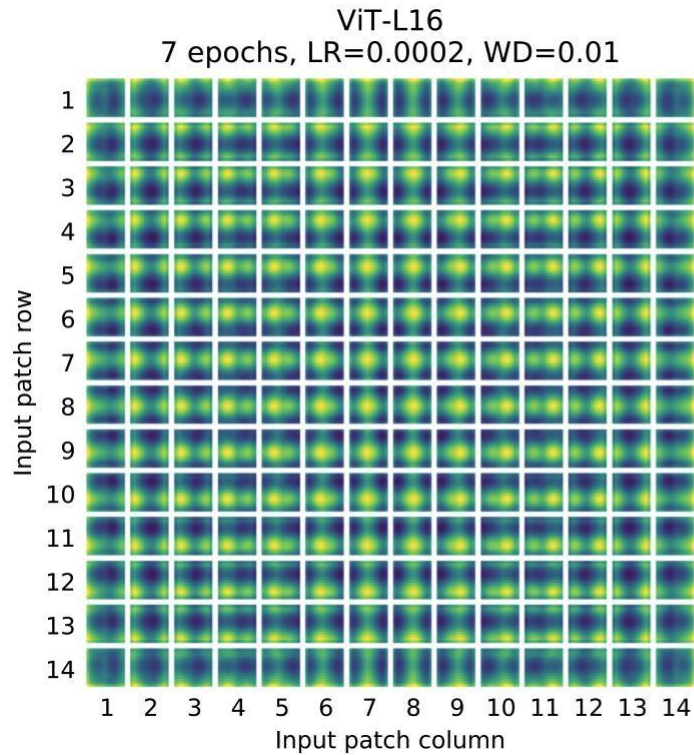


Dosovitskiy, Alexey, et al. "An image is worth 16x16 words: Transformers for image recognition at scale." *arXiv* (2020).

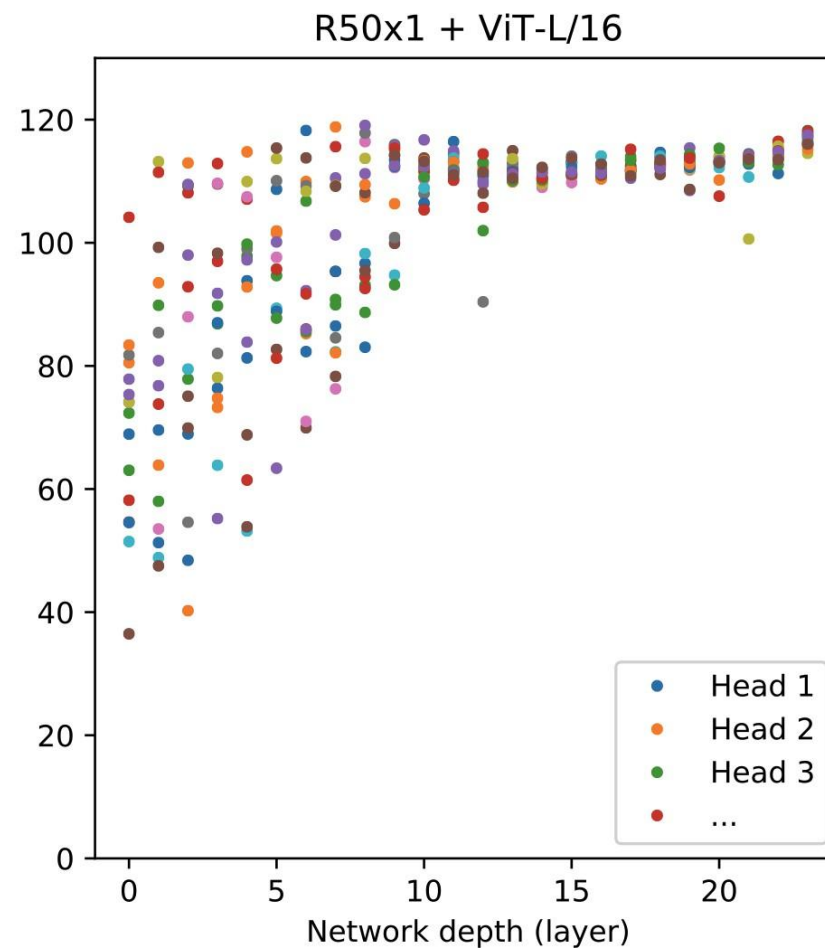
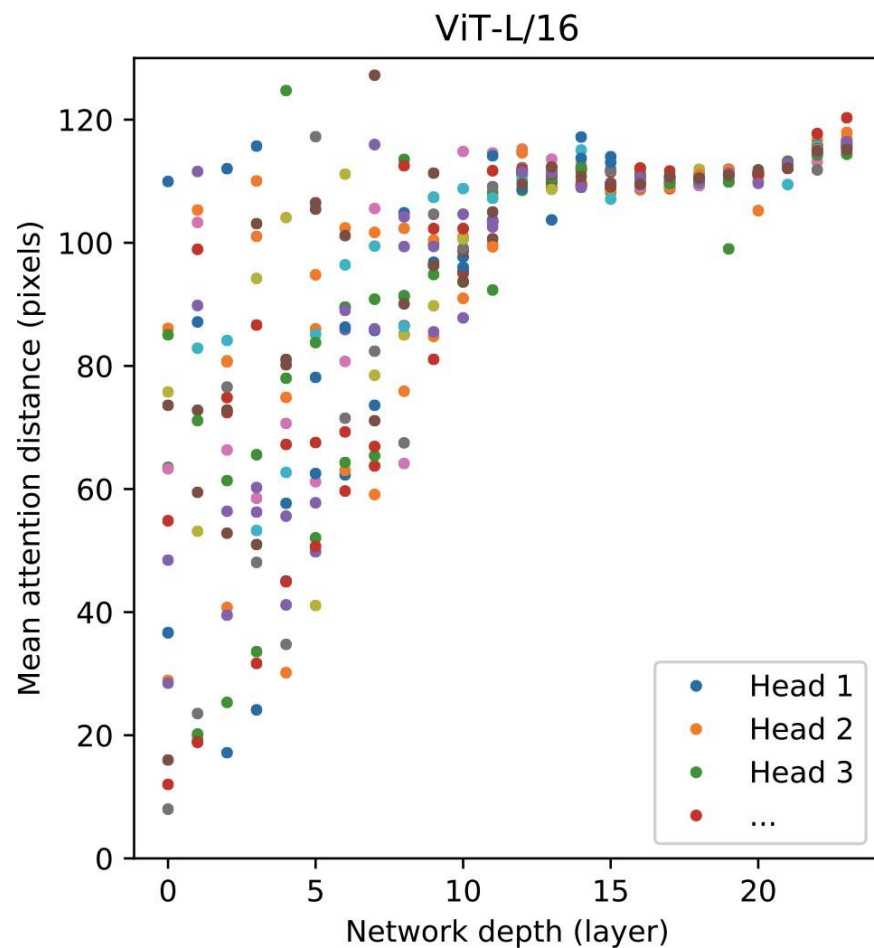
Filters



Learning Location



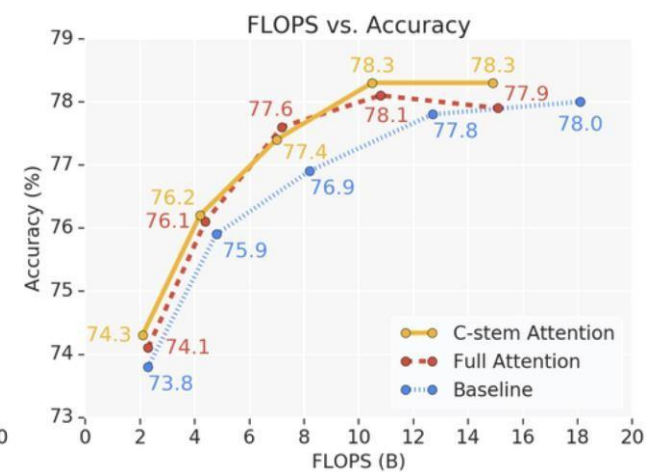
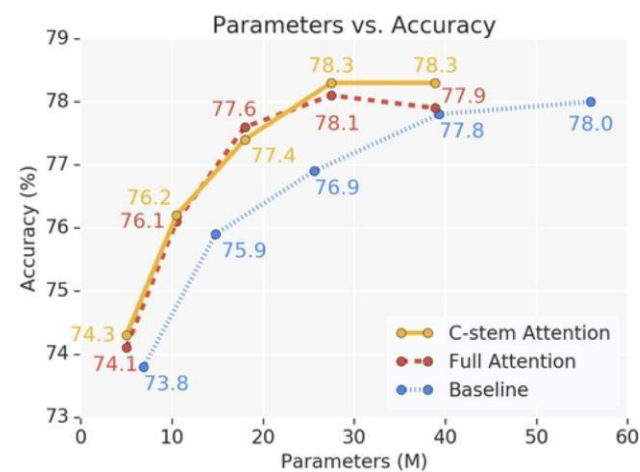
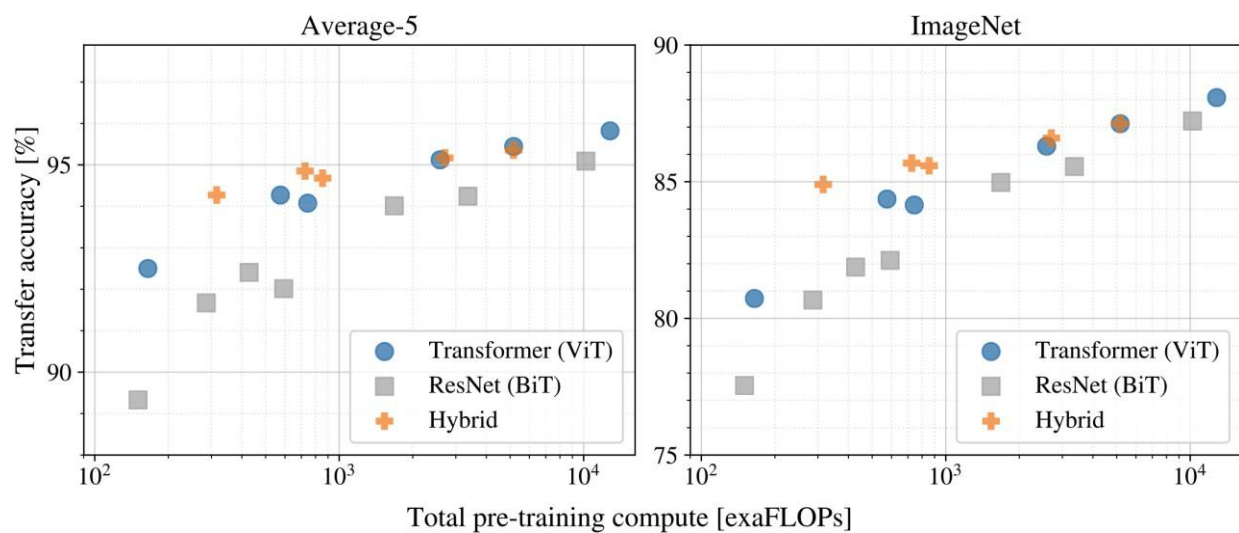
Learning Location



Which is the best?

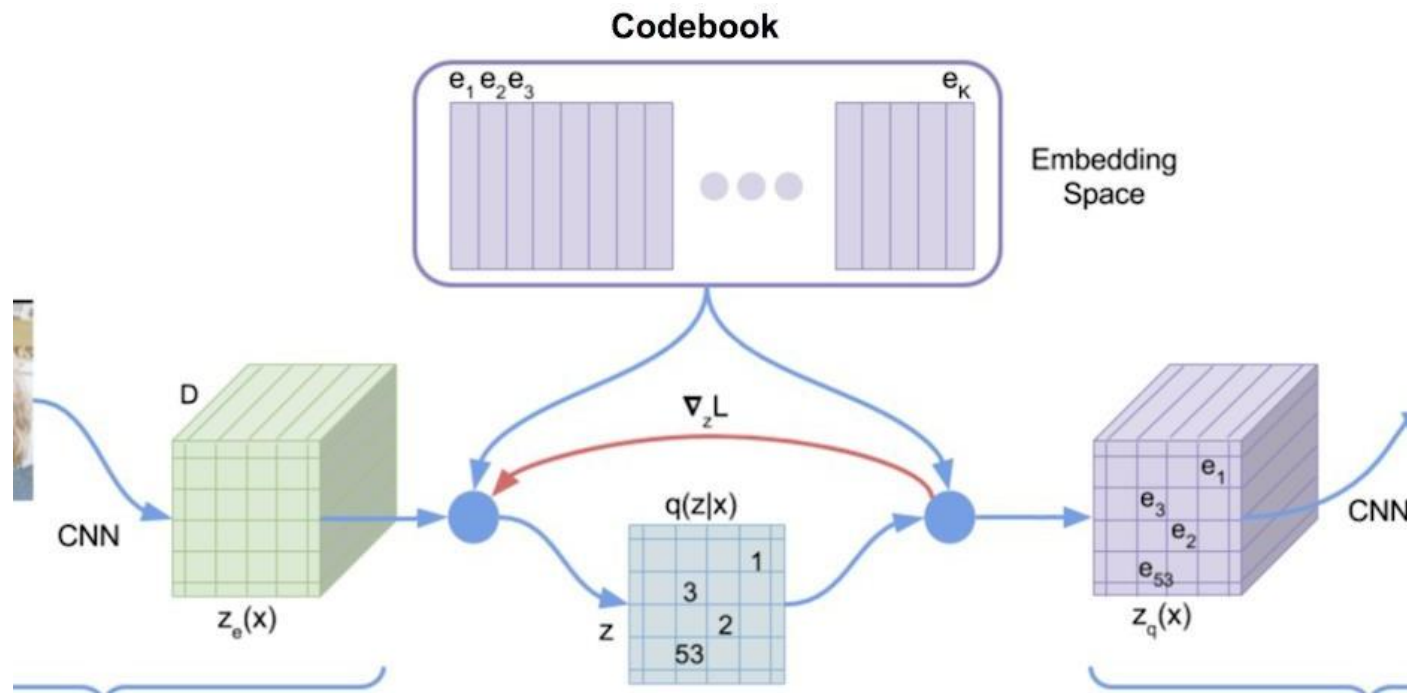
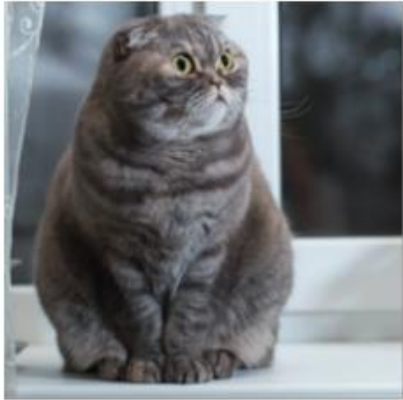
ImageNet	ImageNet ReaL	CIFAR-10	CIFAR-100	Pets	Flowers	exaFLOPs
80.73	86.27	98.61	90.49	93.40	99.27	164
84.15	88.85	99.00	91.87	95.80	99.56	743
84.37	88.28	99.19	92.52	95.83	99.45	574
86.30	89.43	99.38	93.46	96.81	99.66	2586
87.12	89.99	99.38	94.04	97.11	99.56	5172
88.08	90.36	99.50	94.71	97.11	99.71	12826
77.54	84.56	97.67	86.07	91.11	94.26	150
82.12	87.94	98.29	89.20	93.43	97.02	592
80.67	87.07	98.48	89.17	94.08	95.95	285
81.88	87.96	98.82	90.22	94.17	96.94	427
84.97	89.69	99.06	92.05	95.37	98.62	1681
85.56	89.89	99.24	91.92	95.75	98.75	3362
87.22	90.15	99.34	93.53	96.32	99.04	10212
84.90	89.15	99.01	92.24	95.75	99.46	315
85.58	89.65	99.14	92.63	96.65	99.40	855
85.68	89.04	99.24	92.93	96.97	99.43	725
86.60	89.72	99.18	93.64	97.03	99.40	2704
87.12	89.76	99.31	93.89	97.36	99.11	5165

Curves



Visual Tokens

DALL-E's Discrete Variational Autoencoder

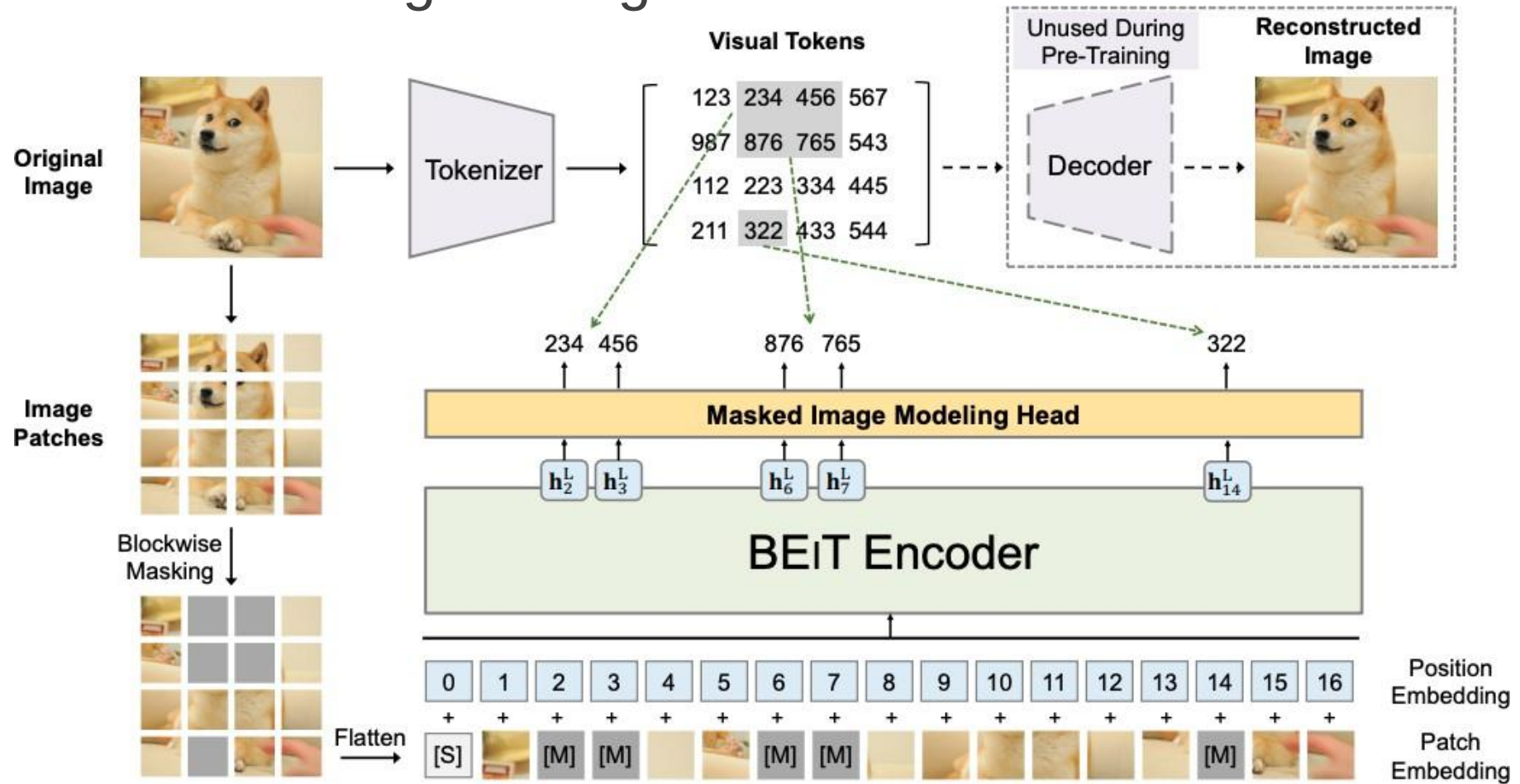


32 x 32 grid of digits, [0... 8192]
Each digit is a “visual token”



Visual Tokens

BeIT: BERT Pre-Training of Image Transformers



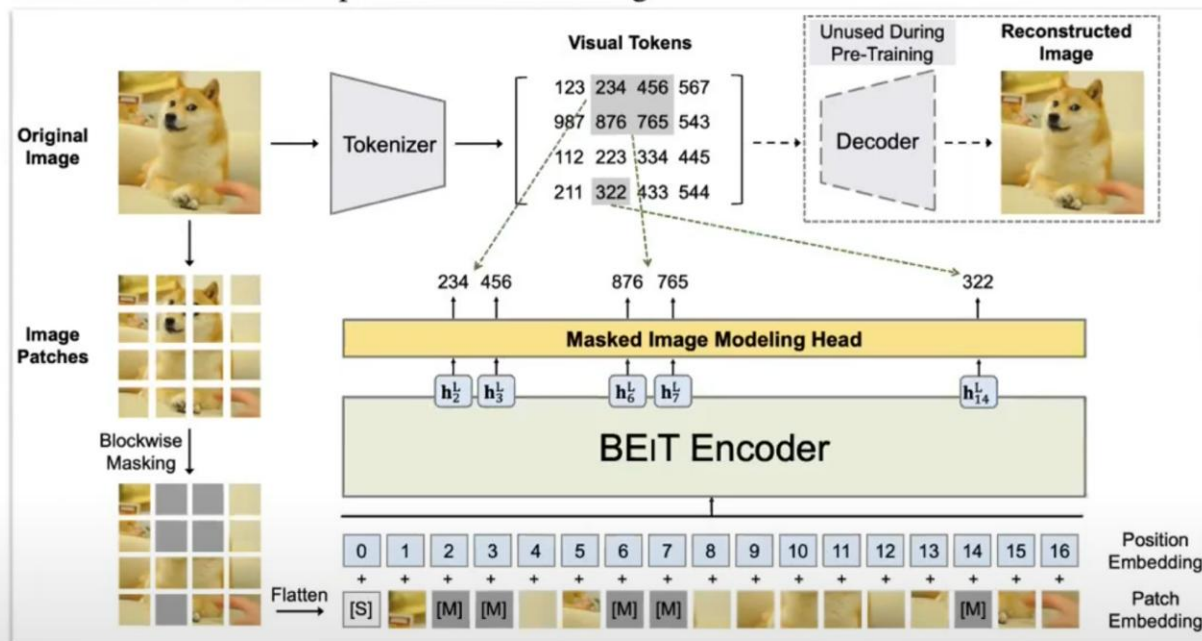
Visual Tokens

BeiT: BERT Pre-Training of Image Transformers



BEiT: BERT Pre-Training of Image Transformers

Bidirectional Encoder representation from Image Transformers



Visual Token

image tokenizer (discrete variational autoencoder)

tokenize $x \in \mathbb{R}^{H \times W \times C}$ into $z \in \mathcal{V}^{h \times w}$

$\mathcal{V} = \{1, \dots, |\mathcal{V}|\} \rightarrow$ vocabulary containing discrete token indices

<https://github.com/openai/DALL-E>

$q_\phi(z|x) \rightarrow$ tokenizer (maps image x into discrete tokens z according to a visual codebook/vocabulary)

Bao, Hangbo, Li Dong, and Furu Wei. "BEiT: BERT Pre-Training of Image Transformers." *arXiv preprint arXiv:2106.08254* (2021).

$p_\psi(x|z) \rightarrow$ decoder (reconstruct the input image x based on the visual tokens z)

$$\max_{\phi, \psi} \mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\psi(x|z)] - \text{KL}[q_\phi(z|x) \parallel p(z)]$$

Gumbel-softmax relaxation

Algorithm 1 Blockwise Masking

Input: $N (= h \times w)$ image patches

Output: Masked positions $\mathcal{M}$

$\mathcal{M} \leftarrow \{\}$

repeat

$s \leftarrow \text{Rand}(16, 0.4N - |\mathcal{M}|)$ ▷ Block size

$r \leftarrow \text{Rand}(0.3, \frac{1}{0.3})$ ▷ Aspect ratio of block

$a \leftarrow \sqrt{s \cdot r}; b \leftarrow \sqrt{s/r}$

$t \leftarrow \text{Rand}(0, h - a); l \leftarrow \text{Rand}(0, w - b)$

$\mathcal{M} \leftarrow \mathcal{M} \cup \{(i, j) : i \in [t, t + a), j \in [l, l + b)\}$

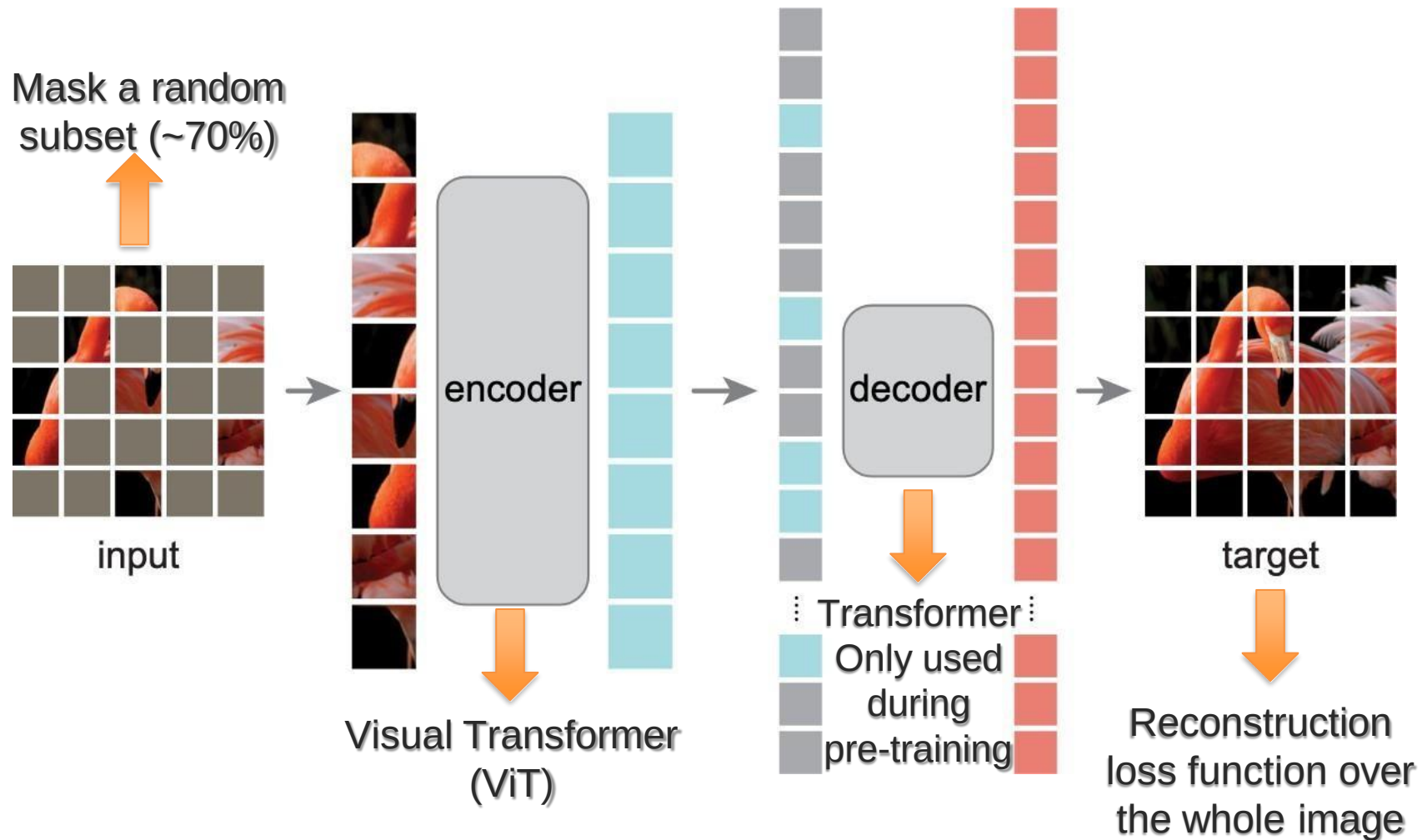
until $|\mathcal{M}| > 0.4N$ ▷ Masking ratio is 40%

return $\mathcal{M}$

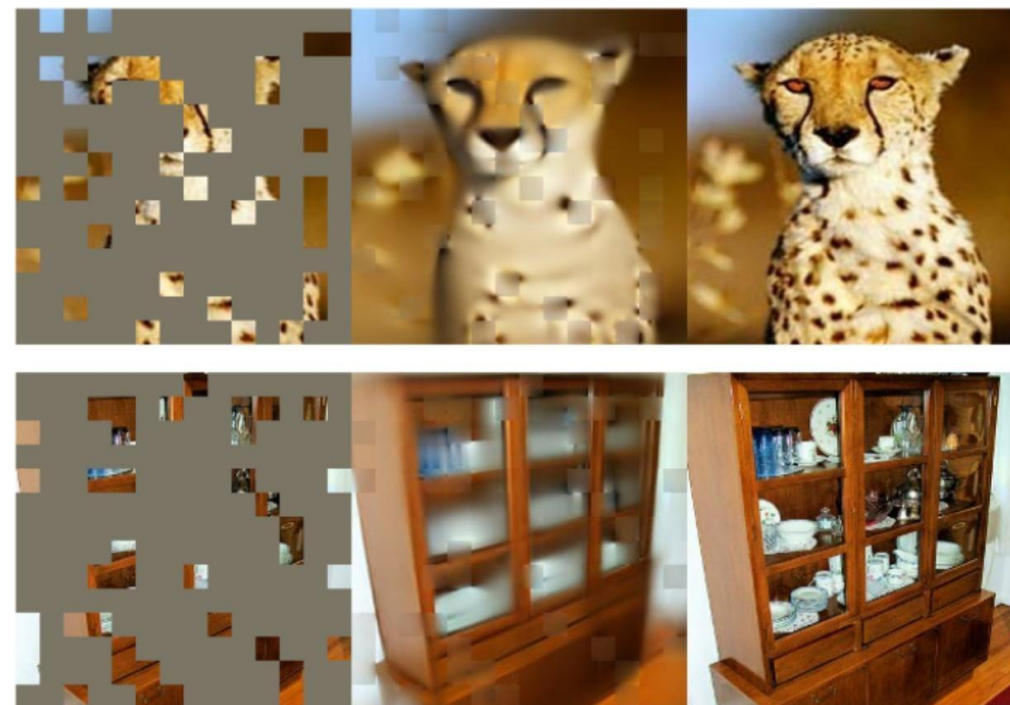
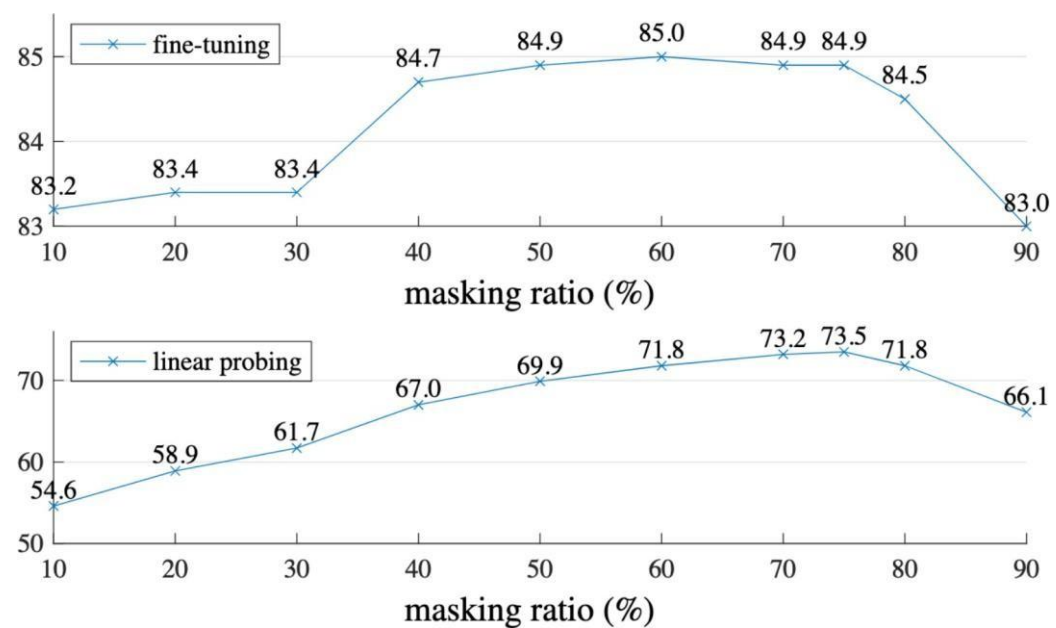
Models	CIFAR-100	ImageNet
<i>Training from scratch (i.e., random initialization)</i>		
ViT ₃₈₄ (Dosovitskiy et al., 2020)	48.5*	77.9
DeiT (Touvron et al., 2020)	n/a	81.8
<i>Supervised Pre-Training on ImageNet-1K (using labeled data)</i>		
ViT ₃₈₄ (Dosovitskiy et al., 2020)	87.1	77.9
DeiT (Touvron et al., 2020)	90.8	81.8
<i>Self-Supervised Pre-Training on ImageNet-1K (without labeled data)</i>		
iGPT-1.36B [†] (Chen et al., 2020a)	n/a	66.5
ViT ₃₈₄ -JFT300M [‡] (Dosovitskiy et al., 2020)	n/a	79.9
DINO (Caron et al., 2021)	91.7	82.8
MoCo v3 (Chen et al., 2021)	87.1	n/a
BEiT (ours)	90.1	83.2
<i>Self-Supervised Pre-Training, and Intermediate Fine-Tuning on ImageNet-1K</i>		
BEiT (ours)	91.8	83.2

Top-1 accuracy of image classification

Masked Auto-Encoder (MAE)

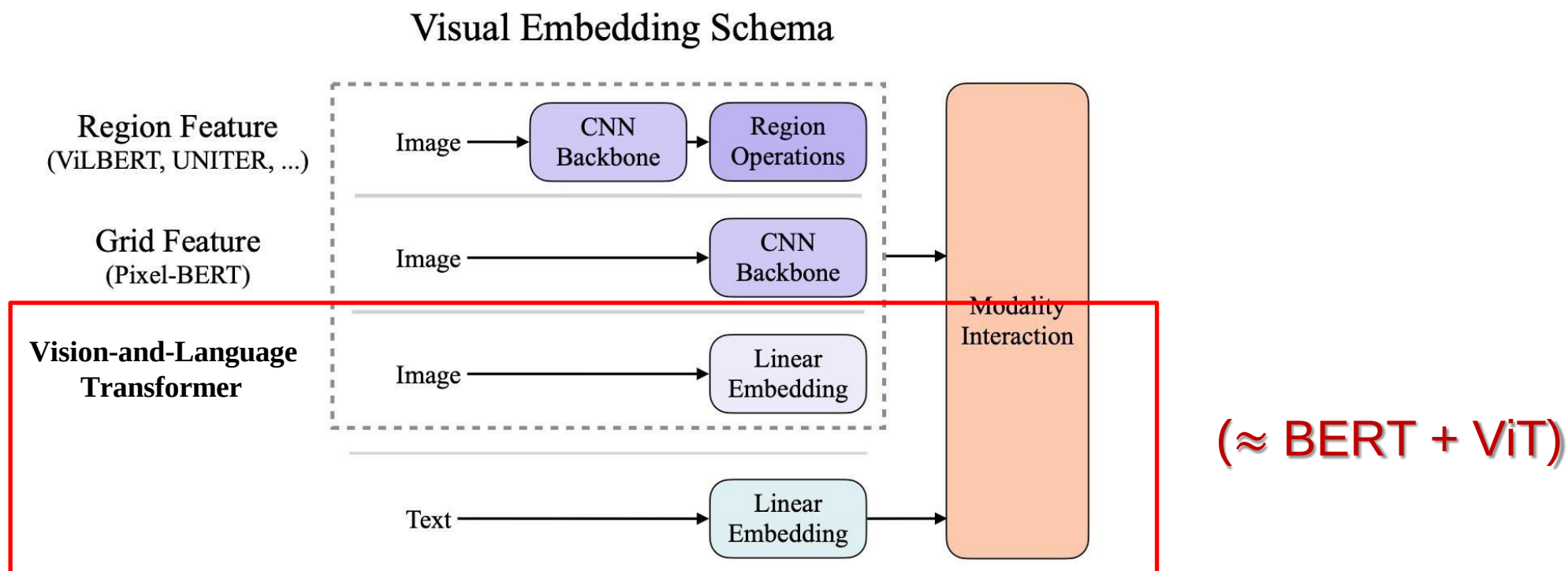


Masked Auto-Encoder (MAE)

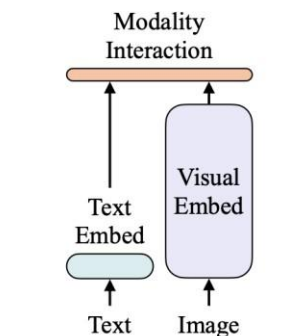
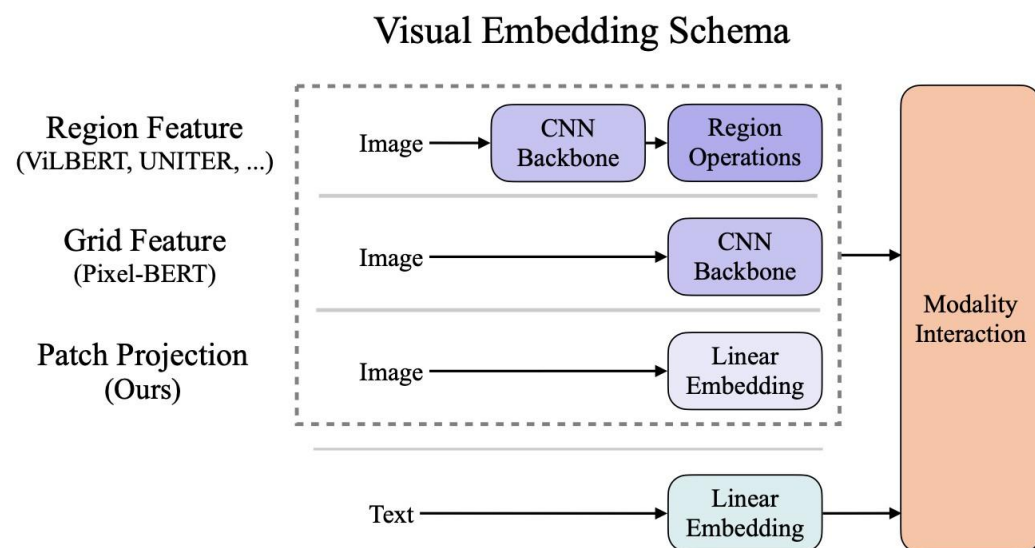


Vision-Language Transformers

Visual Transformers for Multimodal Learning

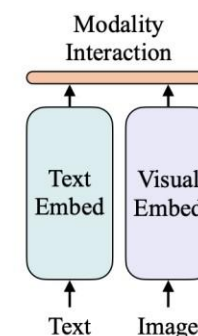


Visual Transformers for Multimodal Learning

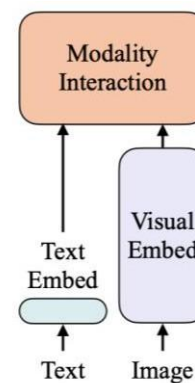


(a) $VE > TE > MI$

e.g. CLIP

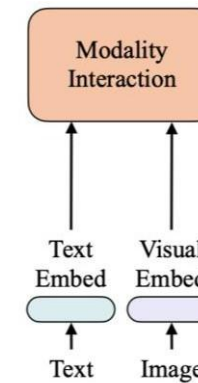


(b) $VE = TE > MI$



(c) $VE > MI > TE$

e.g. LXMERT

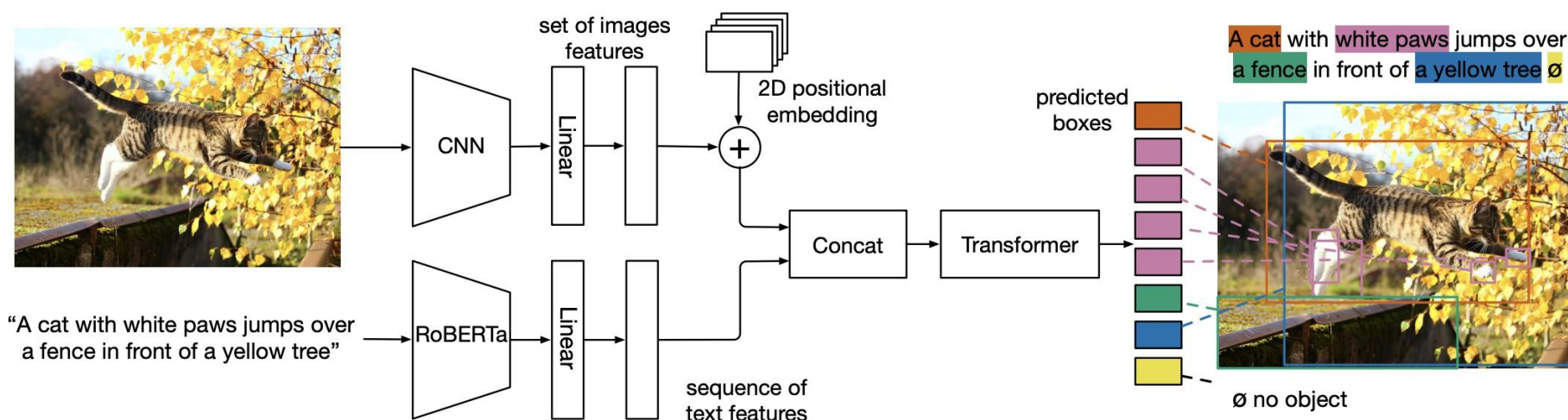
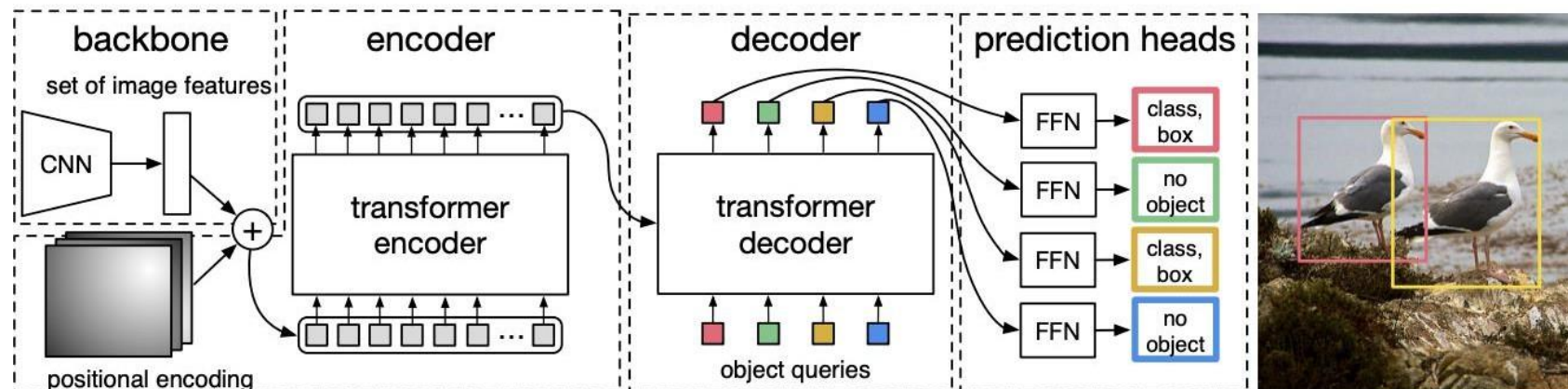


(d) $MI > VE = TE$

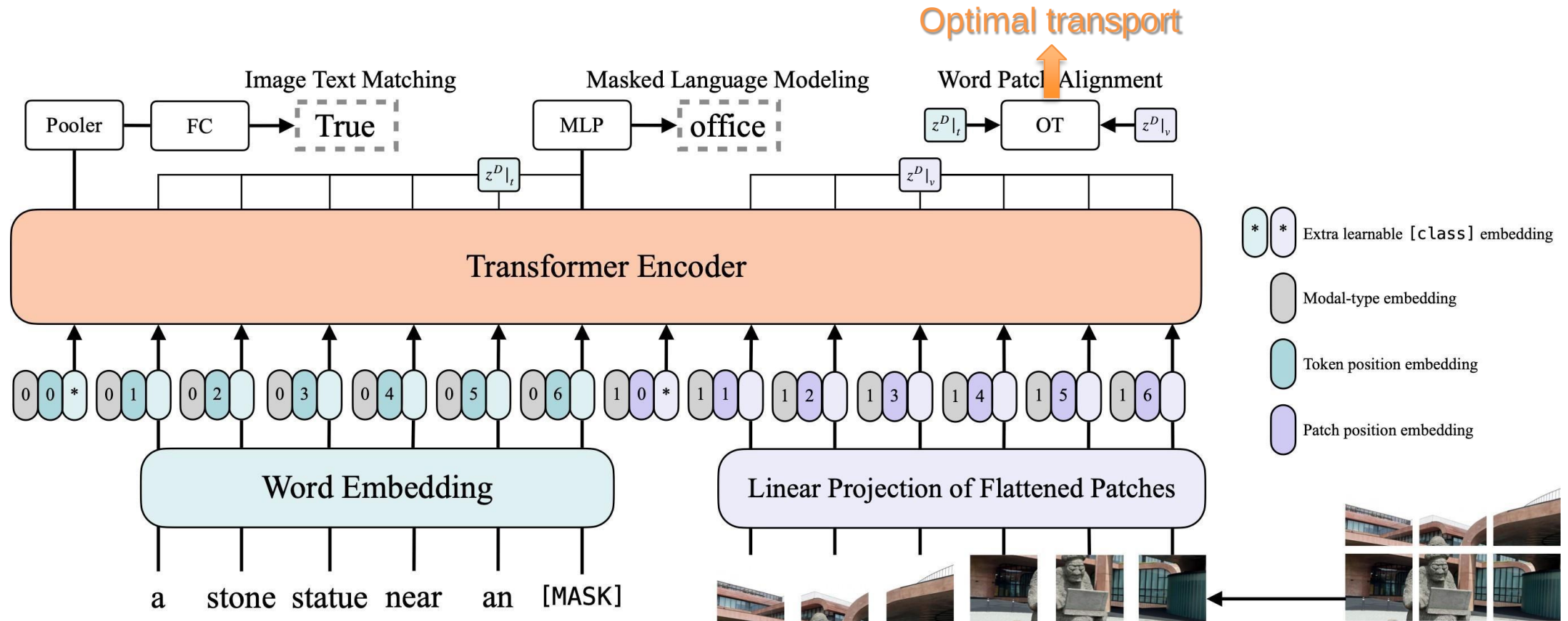
ViLT

DETR / MDETR (CNN+BERT)

Predicting bounding boxes from images (and text)



Visual-and-Language Transformer (ViLT) ($\approx$ BERT + ViT)



Visual-and-Language Transformer (ViLT)

Example of alignment between modalities:



a display of **flowers** growing out and over the retaining **wall** in front of **cottages** on a **cloudy** day.



flowers



wall



cottages



cloudy



a room with a **rug**, a **chair**, a **painting**, and a **plant**.



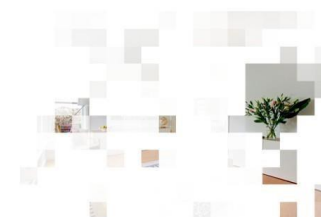
rug



chair



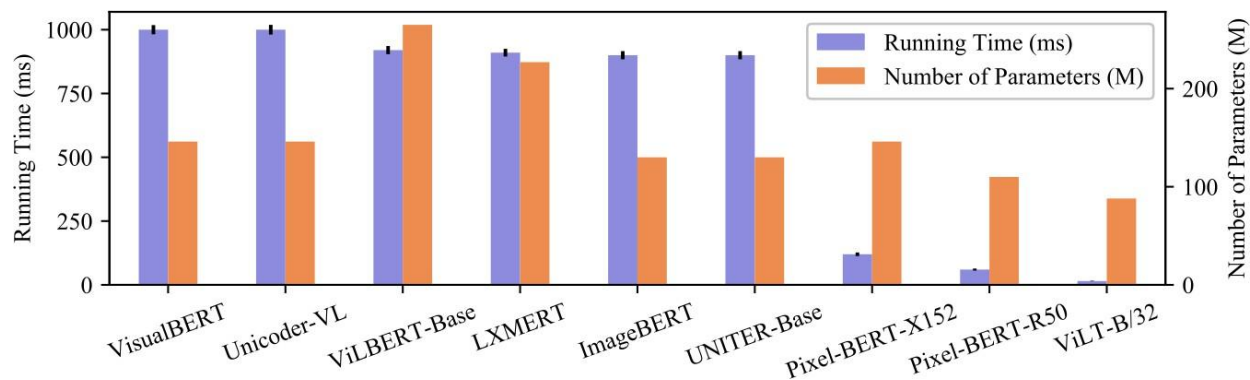
painting



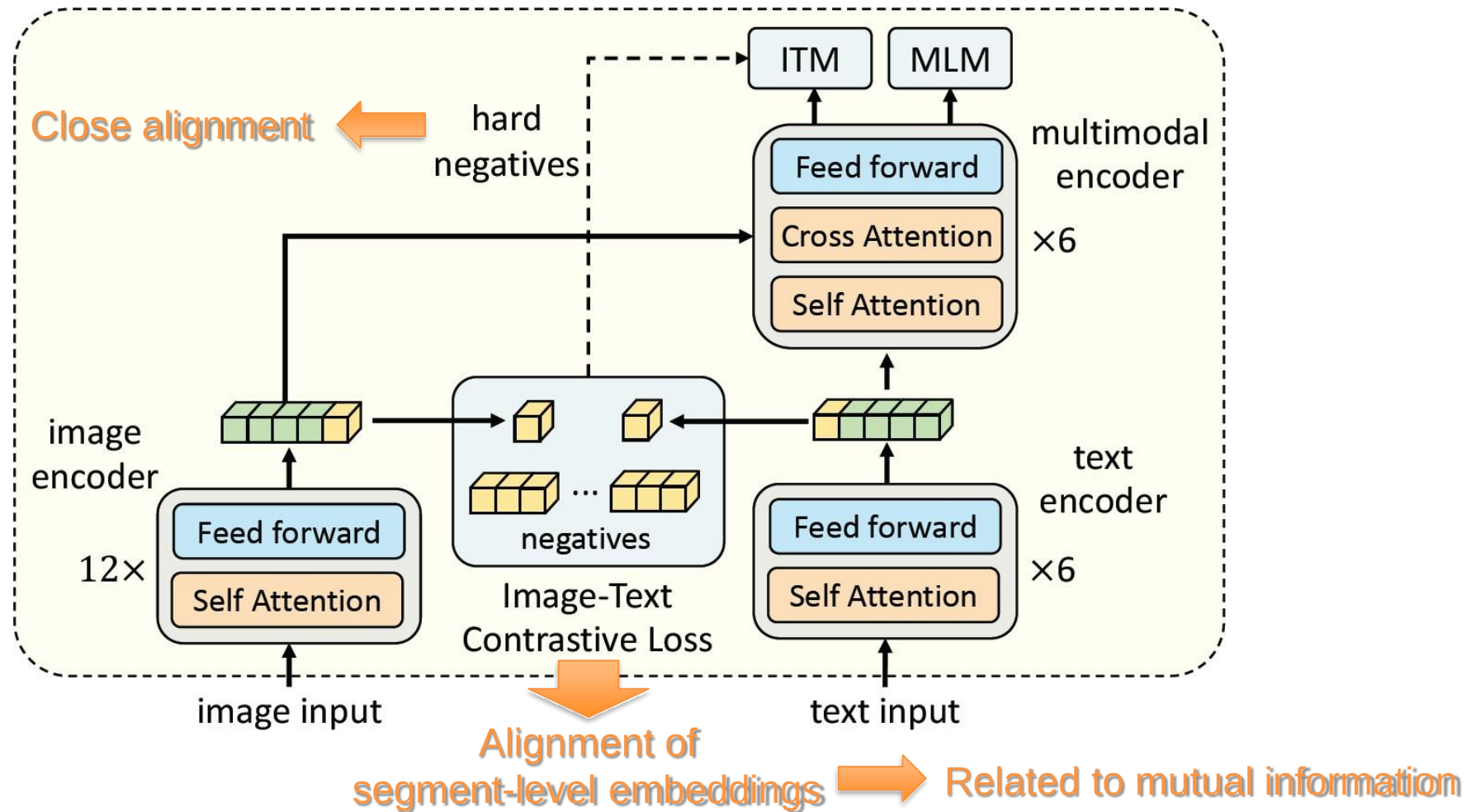
plant

ViLT: Faster Inference?

Visual Embed	Model	Time (ms)	VQAv2 test-dev	NLVR2 dev	NLVR2 test-P
Region	w/o VLP SOTA	~900	70.63	54.80	53.50
	ViLBERT	~920	70.55	-	-
	VisualBERT-Base	~1000	70.80	67.40	67.00
	LXMERT	~910	72.42	74.90	74.50
	UNITER-Base	~900	72.70	75.85	75.80
	OSCAR-Base [†]	~900	73.16	78.07	78.36
	VinVL-Base ^{†‡}	~1000	75.95	82.05	83.08
Grid	Pixel-BERT-X152	~120	74.45	76.50	77.20
	Pixel-BERT-R50	~60	71.35	71.70	72.40
Linear	ViLT-B/32	~15	70.34	74.56	74.66
	ViLT-B/32 [Ⓐ]	~15	70.94	75.24	76.21

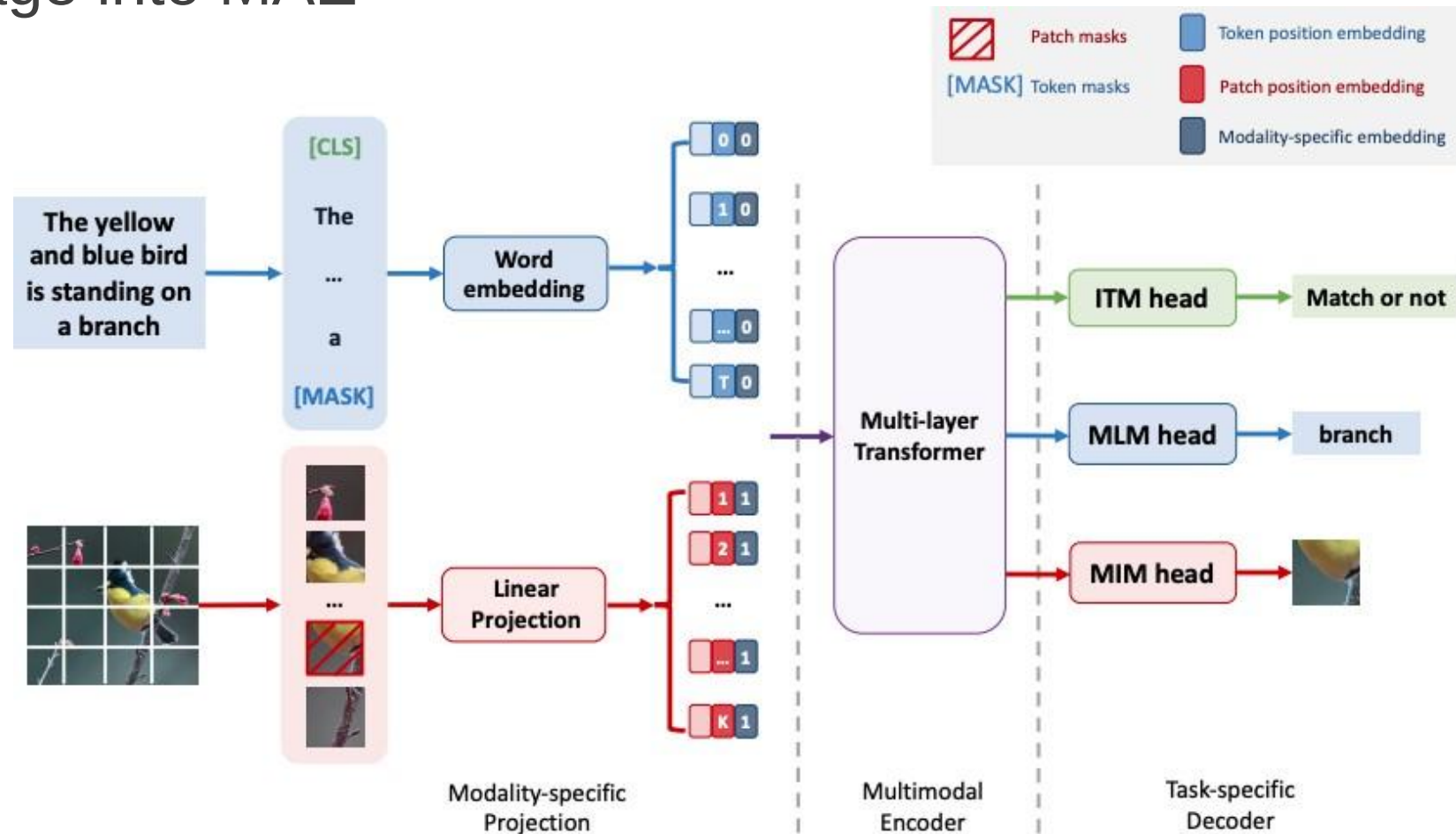


ALBEF: Align Before Fusion ($\approx$ BERT + ViT + CLIP-ish)



Vision-Language from Captions (VLC)

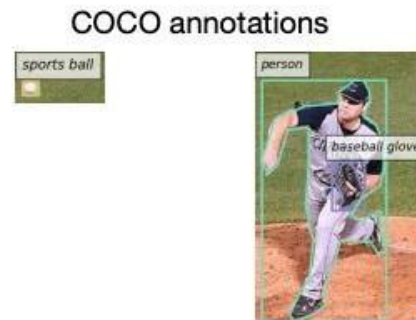
Add language into MAE



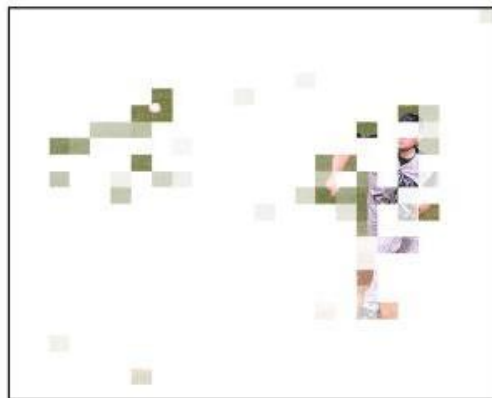
Vision-Language from Captions (VLC)

What are we learning?

*A pitcher at a baseball game who has just **thrown** the ball.*



ViLT (supervised with ImageNet)



Ours (no BBox/class supervision)

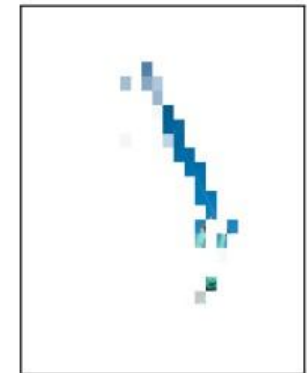
Caption with **focus**

Original Image

ViLT

VLC

A person on a beach holding a kite **string** and a kite is in the air



Vision-Language from Captions (VLC)

What are we learning?

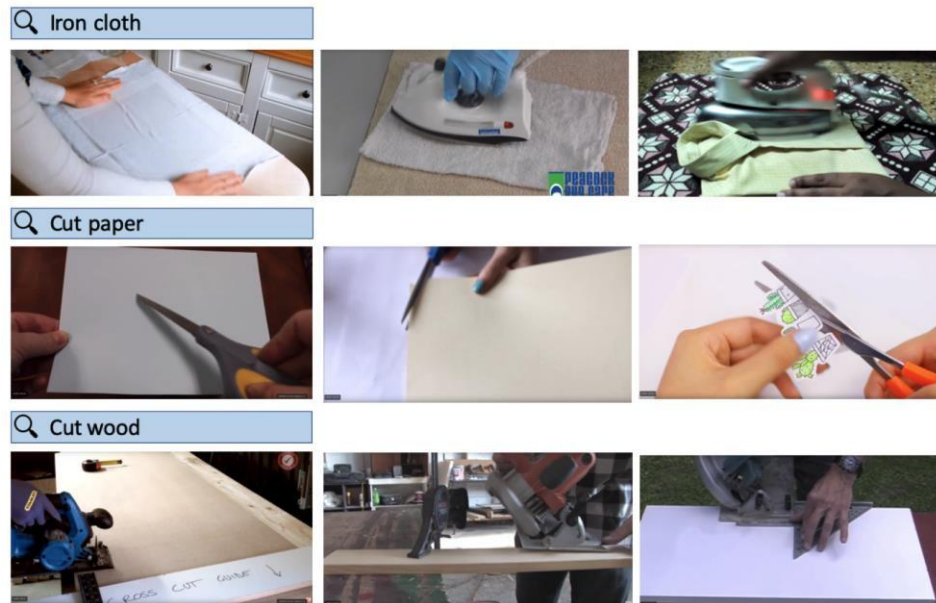
Model	Image Retrieval						
	Flickr30K (1K)			MSCOCO (5K)			
	Params	@1	@5	@10	@1	@5	@10
ViLT [23]	86M	64.4	88.7	93.8	42.7	72.9	83.1
VLC-Base (ours – 5.6M)	86M	72.4	93.4	96.5	50.7	78.9	88.0

Model	VQA _{v2}			NLVR ²	
	Params	test-dev	test-std	dev	test
ViLT [23]	86M	71.26	-	75.70	76.13
<i>No supervised classes or bounding boxes</i>					
VLC-Base (ours – 4M)	86M	72.98	73.03	77.04	78.51

Video Transformers

Video-based Representation and Alignment

HowTo100M benchmark dataset



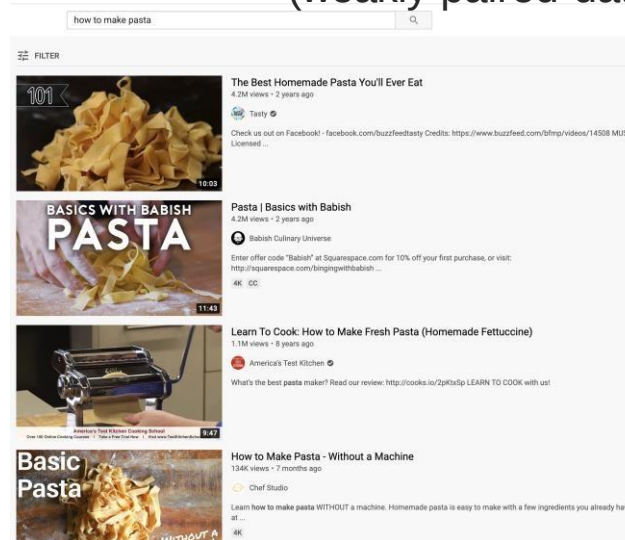
Category	Tasks	Videos	Clips
Food and Entertaining	11504	497k	54.4M
Home and Garden	5068	270k	29.5M
Hobbies and Crafts	4273	251k	29.8M
Cars & Other Vehicles	810	68k	7.8M
Pets and Animals	552	31k	3.5M
Holidays and Traditions	411	27k	3.0M
Personal Care and Style	181	16k	1.6M
Sports and Fitness	205	16k	2.0M
Health	172	15k	1.7M
Education and Communications	239	15k	1.6M
Arts and Entertainment	138	10k	1.2M
Computers and Electronics	58	5k	0.6M
Total	23.6k	1.22M	136.6M

Visual Representations from Uncurated Instructional Videos

Goal: Learn better visual representations...

... by taking advantage of large-scale video+language resources

Instructional videos
(weakly-paired data)



it's turning into a much thicker mixture



The biggest mistake is not kneading it enough



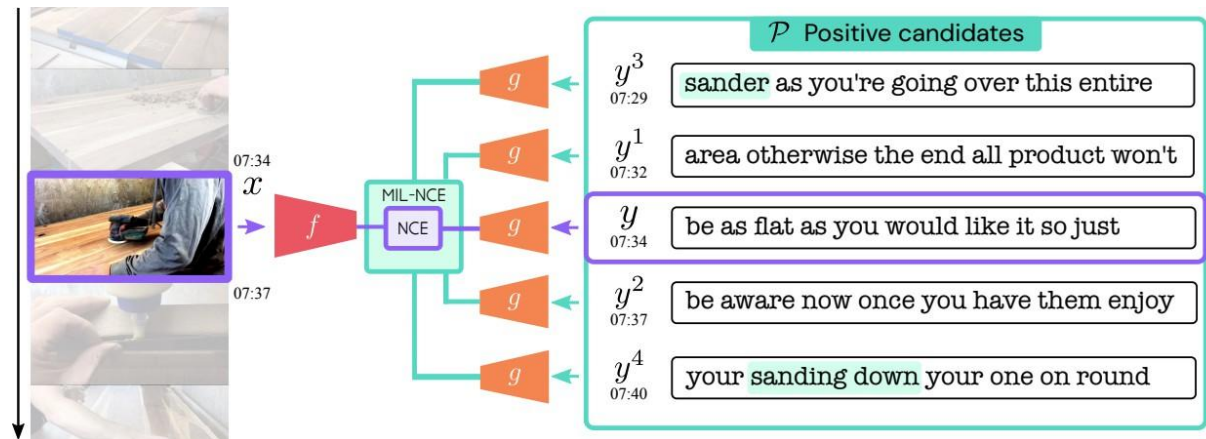
...

End-to-End Learning of Visual Representations from Uncurated Instructional Videos

Antoine Miech, Jean-Baptiste Alayrac, Lucas Smaira, Ivan Laptev, Josef Sivic, and Andrew Zisserman – CVPR 2020

Weakly Paired Data

Data point: “a short 3.2 seconds video clip (32 frames at 10 FPS) together with a small number of words (not exceeding 16)”



How to handle this misalignment?

Multi-instance learning!

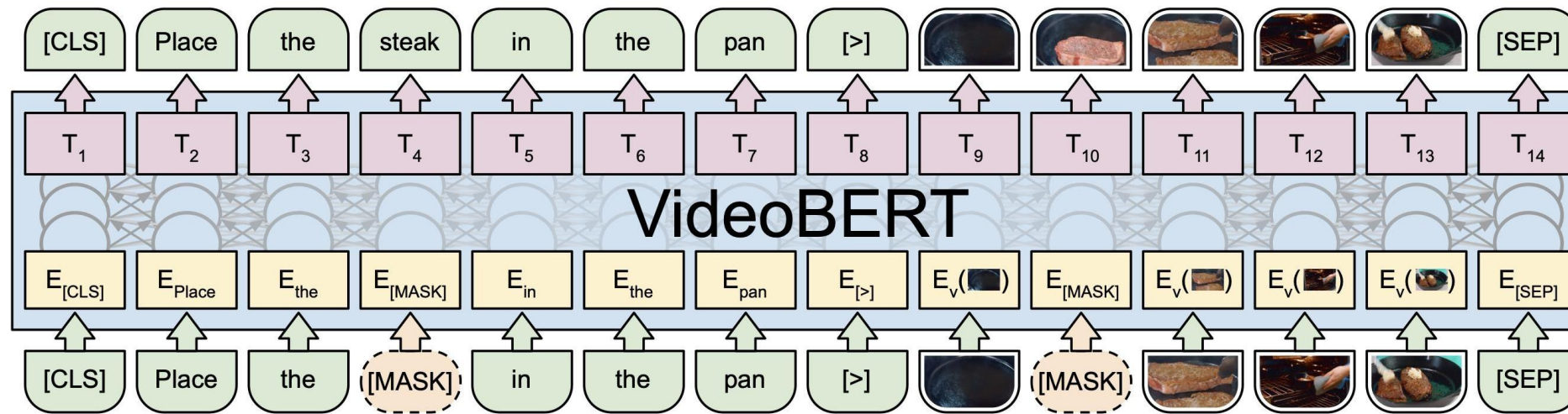
How to do it self-supervised?

Contrastive learning!

End-to-End Learning of Visual Representations from Uncurated Instructional Videos

Antoine Miech, Jean-Baptiste Alayrac, Lucas Smaira, Ivan Laptev, Josef Sivic, and Andrew Zisserman – CVPR 2020

Another Approach for Weakly-Paired Video Data



How do we get visual words now?

**K-mean clustering
+ centroid**

ActBERT

